

2162. Minimum Cost to Set Cooking Time

Solved ●

Medium Topics Hint

A generic microwave supports cooking times for:

- at least 1 second.
- at most 99 minutes and 99 seconds.

To set the cooking time, you push **at most four digits**. The microwave normalizes what you push as four digits by **prepending zeroes**. It interprets the **first** two digits as the minutes and the **last** two digits as the seconds. It then **adds** them up as the cooking time. For example,

- You push 9 5 4 (three digits). It is normalized as 0954 and interpreted as 9 minutes and 54 seconds.
- You push 0 0 0 8 (four digits). It is interpreted as 0 minutes and 8 seconds.
- You push 8 0 9 0. It is interpreted as 80 minutes and 90 seconds.
- You push 8 1 3 0. It is interpreted as 81 minutes and 30 seconds.

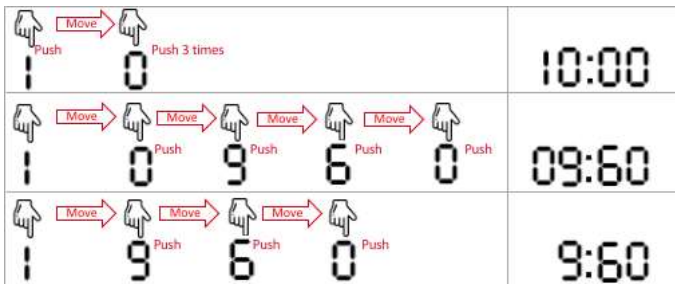
You are given integers `startAt`, `moveCost`, `pushCost`, and `targetSeconds`. **Initially**, your finger is on the digit `startAt`. Moving the finger above **any specific digit** costs `moveCost` units of fatigue. Pushing the digit below the finger **once** costs `pushCost` units of fatigue.

There can be multiple ways to set the microwave to cook for `targetSeconds` seconds but you are interested in the way with the minimum cost.

Return the **minimum cost** to set `targetSeconds` seconds of cooking time.

Remember that one minute consists of 60 seconds.

Example 1:



Input: `startAt = 1`, `moveCost = 2`, `pushCost = 1`, `targetSeconds = 600`

Output: 6

Explanation: The following are the possible ways to set the cooking time.

- 1 0 0 0, interpreted as 10 minutes and 0 seconds.

The finger is already on digit 1, pushes 1 (with cost 1), moves to 0 (with cost 2), pushes 0 (with cost 1), pushes 0 (with cost 1), and pushes 0 (with cost 1).

The cost is: $1 + 2 + 1 + 1 + 1 = 6$. This is the minimum cost.

- 0 9 6 0, interpreted as 9 minutes and 60 seconds. That is also 600 seconds.

The finger moves to 0 (with cost 2), pushes 0 (with cost 1), moves to 9 (with cost 2), pushes 9 (with cost 1), moves to 6 (with cost 2), pushes 6 (with cost 1), moves to 0 (with cost 2), and pushes 0 (with cost 1).

The cost is: $2 + 1 + 2 + 1 + 2 + 1 + 2 + 1 = 12$.

- 9 6 0, normalized as 0960 and interpreted as 9 minutes and 60 seconds.

The finger moves to 9 (with cost 2), pushes 9 (with cost 1), moves to 6 (with cost 2), pushes 6 (with cost 1), moves to 0 (with cost 2), and pushes 0 (with cost 1).

The cost is: $2 + 1 + 2 + 1 + 2 + 1 = 9$.

Example 2:



Input: `startAt = 0`, `moveCost = 1`, `pushCost = 2`, `targetSeconds = 76`

Output: 6

Explanation: The optimal way is to push two digits: 7 6, interpreted as 76 seconds.

The finger moves to 7 (with cost 1), pushes 7 (with cost 2), moves to 6 (with cost 1), and pushes 6 (with cost 2). The total cost is: $1 + 2 + 1 + 2 = 6$

Note other possible ways are 0076, 076, 0116, and 116, but none of them produces the minimum cost.

Constraints:

- $0 \leq \text{startAt} \leq 9$
- $1 \leq \text{moveCost}, \text{pushCost} \leq 10^5$
- $1 \leq \text{targetSeconds} \leq 6039$

Seen this question in a real interview before? 1/6

Yes No

Accepted 24,245/53.6K | Acceptance Rate 45.2 %

 Topics 

 Hint 1 

 Hint 2 

 Hint 3 

 Hint 4 

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